浅不可变类实现

- 姓名：陈雨萱
- 学号：23300240034
- 专业：计算机科学与技术（拔尖人才计划）

检查情况和实现逻辑

1. 判断类名是否是“_Immutable”类（为检查起辅助作用）
   对于长度有10个字符的类名，获取最后10个字符，判断是否为“_Immutable”类。
2. 对_Immutable类的继承检查，即在ClassDecl中检查继承一致性
   对于当前的node，获取它的父类，调用上述工具函数分别判断父类和它到底是不是“_Immutable”类，如果一个是一个不是，就输出报错信息，不满足继承一致性。
3. 判断是否对它进行了改变，即在Assign中禁止将它置为lvalue
   在Assign中，对于符号左边的object，先获取它的类型，如果是class，调用工具函数判断它是否为“_Immutable”类，如果是，则输出报错信息，因为该类变量本身不允许改变。但是止步于此，不分析类的成员。

测试结果

我的实现是，如果有语义错误，找出所有错误，不生成semant。
新写的两个案例主要考虑，对于浅不可变类的类变量，它指向的成员是可以被改变的(itest)；以及继承来的类变量不允许被改写(itest2)。
immutabletest1的输出：

```
root@cyx:~/fducompilerh2026/fducompilerh2026/HW2/build# ./tools/main/main --debug ../test/immutabletest1
Read AST from file: ../test/immutabletest1.2.ast
Converting XML to AST...
Semantic analysis...
--Making Name Maps...
Classes: A_Immutable ; TestClass_Immutable ; __$main__ ;
Class Hiearchy: A_Immutable->TestClass_Immutable ;
Methods: TestClass_Immutable->test1 ; __$main__->main ;
Class Variables: A_Immutable->a with type=Int ; A_Immutable->o with type=Class ; TestClass_Immutable->a with type=Int ; TestClass_Immutable->b with type=IntArray ;
Method Variables: TestClass_Immutable->test1->a with type=IntArray ; __$main__->main->a with type=Class ;
Method Formals: TestClass_Immutable->test1->_^return^_test1 with type=Class ; TestClass_Immutable->test1->a with type=Class ; __$main__->main->_^return^_main with type=Int ;
--Analyzing Semantics...
Start Semantic Analysis
Error: Cannot assign to field of immutable class A_Immutable at Position(sline: 3, scolumn: 5, eline: 3, ecolumn: 21)
Error: Cannot assign to field of immutable class A_Immutable at Position(sline: 4, scolumn: 5, eline: 4, ecolumn: 36)
Semantic Analysis failed due to errors.
root@cyx:~/fducompilerh2026/fducompilerh2026/HW2/build#
```

immutabletest2的输出：

```
root@cyx:~/fducompilerh2026/fducompilerh2026/HW2/build# ./tools/main/main --debug ../test/immutabletest2
Read AST from file: ../test/immutabletest2.2.ast
Converting XML to AST...
Semantic analysis...
--Making Name Maps...
Classes: A_Immutable ; B ; TestClass_Immutable ; __$main__ ;
Class Hiearchy: A_Immutable->TestClass_Immutable ; B->TestClass_Immutable ;
Methods: __$main__->main ;
Class Variables: A_Immutable->a with type=Int ; A_Immutable->o with type=Class ; B->i with type=Int ; TestClass_Immutable->a with type=Int ; TestClass_Immutable->b with type=IntArray ;
Method Variables:
Method Formals: __$main__->main->_^return^_main with type=Int ;
--Analyzing Semantics...
Start Semantic Analysis
Error: Immutable class inheritance violation at Position(sline: 14, scolumn: 1, eline: 16, ecolumn: 1)
Semantic Analysis failed due to errors.
```

itest的输出：

```
root@cyx:~/fducompilerh2026/fducompilerh2026/HW2/build# ./tools/main/main --debug ../test/itest
Read AST from file: ../test/itest.2.ast
Converting XML to AST...
Semantic analysis...
--Making Name Maps...
Classes: A_Immutable ; __$main__ ;
Class Hiearchy:
Methods: __$main__->main ;
Class Variables: A_Immutable->x with type=IntArray ;
Method Variables: __$main__->main->a with type=Class ;
Method Formals: __$main__->main->_^return^_main with type=Int ;
--Analyzing Semantics...
Start Semantic Analysis
Semantic Analysis Done
Convert AST to XML with Semantic Info...
```

itest2的输出：

```
root@cyx:~/fducompilerh2026/fducompilerh2026/HW2/build# ./tools/main/main --debug ../test/itest2
Read AST from file: ../test/itest2.2.ast
Converting XML to AST...
Semantic analysis...
--Making Name Maps...
Classes: A ; TestClass_Immutable ; __$main__ ;
Class Hiearchy: A->TestClass_Immutable ;
Methods: __$main__->main ;
Class Variables: A->a with type=Int ; A->o with type=Class ; TestClass_Immutable->a with type=Int ; TestClass_Immutable->b with type=IntArray ;
Method Variables: __$main__->main->a with type=Class ;
Method Formals: __$main__->main->_^return^_main with type=Int ;
--Analyzing Semantics...
Start Semantic Analysis
Error: Immutable class inheritance violation at Position(sline: 11, scolumn: 1, eline: 14, ecolumn: 1)
Semantic Analysis failed due to errors.
```